Lecture #15: The Central Limit Theorem

STAT 200 – Introductory Statistics

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The Central Limit Theorem

Let X be a random variable with mean μ and finite variance σ^2. Let us draw a random sample of size n from this distribution.
Then, the distribution of the sample sum, $T = \sum_{i=1}^{n} X_i$, gets closer and closer to a Normal distribution as n gets larger. Specifically, for large n,

$$T = \sum_{i=1}^{n} X_i \mathrel{\dot\sim} \text{Normal}(n\mu, n\sigma^2).$$

(The symbol $\dot\sim$ means "is approximately distributed as".)
The proof of this theorem is beyond the scope of this course. It is first proven in MATH 321: Mathematical Statistics I.

Consequences of the Central Limit Theorem

The Central Limit Theorem (CLT) tells us the following:

- The *sum* of independent random variables is more Normal than the distribution of the variable itself, unless the variable is Normally distributed or has an infinite variance.
  - The Binomial is a sum of independent Bernoulli random variables.
  - The Poisson is a sum of independent Poisson random variables.
- Because the sample mean is just the sample sum, divided by a constant (n), the Central Limit Theorem tells us that *the distribution of sample means* will tend towards the Normal distribution.
- The speed of convergence depends on how close the data distribution is to a Normal distribution. The closer, the faster.

Example #1: The Uniform Distribution

Let $X \sim \text{Uniform}(a, b)$. Use the Central Limit Theorem to estimate the distribution of the sum of a sample of size n.
By the CLT,

$$T \sim \text{Normal}(n\mu, n\sigma^2).$$

From our knowledge of the Uniform distribution, this means

$$T \sim \text{Normal}\left(n\left(\frac{a+b}{2}\right), n\left(\frac{(b-a)^2}{12}\right)\right)$$

Example #2: The Exponential Distribution

Let $X \sim \text{Exponential}(\lambda)$. Use the Central Limit Theorem to estimate the distribution of the sum of a sample of size n.
By the CLT,

$$T \sim \text{Normal}(n\mu, n\sigma^2).$$

From our knowledge of the Exponential distribution, this means

$$T \sim \text{Normal}\left(n\left(\frac{1}{\lambda}\right), n\left(\frac{1}{\lambda^2}\right)\right)$$

Example #3: The Binomial Distribution

Let $X \sim \text{Binomial}(n, p)$. Use the Central Limit Theorem to approximate the distribution of X.

Note that the Binomial is just the sum of n independent Bernoulli distributions. That is, if $Y_i \sim \text{Bernoulli}(p)$, then $Y_i \sim \text{Binomial}(1, p)$ and,

$$X = \sum_{i=1}^{n} Y_i \sim \text{Binomial}(n, p).$$

Thus, by the Central Limit Theorem, we know

$$X \sim \text{Normal}(np, np(1-p)).$$

The Distribution of the Sample Mean

Let X be a random variable with mean μ and finite variance σ^2. Let us draw a random sample of size n from this distribution.
Then, for large n, the distribution of the sample mean is approximately

$$\overline{X}_n \dot{\sim} \text{Normal}\left(\mu, \frac{1}{n}\sigma^2\right)$$

Proof of the Distribution of the Sample Mean

Proof: From the Central Limit Theorem, we know

$$T = \sum_{i=1}^{n} X_i \dot{\sim} \text{Normal}(n\mu, n\sigma^2).$$

Thus,

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i \sim \text{Normal}\left(\mu, \frac{1}{n}\sigma^2\right)$$

Proof of the Distribution of the Sample Mean

Proof of Expectation:

$$\begin{aligned}\mathbb{E}(\overline{X}_n) &= \mathbb{E}\left(\frac{1}{n}T\right) \\ &= \frac{1}{n}\mathbb{E}(T) \\ &= \frac{1}{n}\left(n\mu\right) \\ &= \mu\end{aligned}$$

Proof of the Distribution of the Sample Mean

Proof of Variance:

$$\begin{aligned}
\mathbb{V}(\overline{X}_n) &= \mathbb{V}\left(\frac{1}{n}T\right) \\
&= \frac{1}{n^2}\mathbb{V}(T) \\
&= \frac{1}{n^2}\left(n\sigma^2\right) \\
&= \frac{1}{n}\sigma^2
\end{aligned}$$

NOTE: Sample Means vs. Sample Values

Before we move onto examples, I want to make an important distinction.
The Central Limit Theorem describes the behavior of the ***sample mean***, not the ***sample values***!
The sample values will stay in whatever distribution the population takes. The sample means will approach Normality.
It is a subtle difference, but a very important one.
`https://www.statcrunch.com/applets/type3&samplingdist`

Example #1: A First Look

I draw a sample of size $n = 14$ from a population with mean $\mathbb{E}(X) = 126$ and variance $\mathbb{V}(X) = 42$. What is the approximate distribution of the sample means?

Example #2: Heights

I have been told that adult height for males in the United States has mean $\mu = 69$ inches and standard deviation $\sigma = 3$ inches. What is the probability of having the mean of a sample of size 2 being less than 65 inches?

Example #3: More Heights

I have been told that adult height for males in the United States has mean $\mu = 69$ inches and standard deviation $\sigma = 3$ inches. What is the probability of having the mean of a sample of size 10 being less than 65 inches?

Example #4: Crime

The 2000 violent crime rates for the 50 states (and the District of Columbia) are given in the data file `crime`. What is a 95% central confidence interval for the mean violent crime rate?

Example #4: Crime (Bootstrapping)

We could also estimate the confidence interval from the data. This process is called "bootstrapping", and here is the code:

```
mn = numeric()
for (i in 1:1e4) {
    x = sample(vcrime00, replace = TRUE)
    mn[i] = mean (x)
}

quantile(mn, c(0.025, 0.975))
```

This gives a 95% confidence interval of 380 to 510.

Question: Why is there a difference between the two confidence intervals?

The Distribution of the Sample Proportion

Let $X \sim \text{Binomial}(n, p)$ be the number of successes in a random sample of n independent Bernoulli(p) random variables.
Then, the distribution of the sample proportion is

$$P = \frac{1}{n}X \dot{\sim} \text{Normal}\left(p, \frac{p(1-p)}{n}\right)$$

Proof of the Distribution of the Sample Proportion

Proof: From the Central Limit Theorem, we know

$$X \sim \text{Normal}(np, np(1-p)).$$

Thus,

$$\mathbb{E}(P) = \mathbb{E}\left(\frac{X}{n}\right) = \frac{\mathbb{E}(X)}{n} = \frac{np}{n} = p,$$
$$\mathbb{V}(P) = \mathbb{V}\left(\frac{X}{n}\right) = \frac{\mathbb{V}(X)}{n^2} = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}.$$

Proof of the Distribution of the Sample Proportion

So we have:

$$P \sim \text{Normal}\left(p, \frac{p(1-p)}{n}\right).$$

Example #1: Poverty

According to the US Census, 18% of Americans are below the poverty line. If I randomly sample $n = 10$ people from the United States, what is the probability that more than 20% of them are below the poverty line?

Example #2: More Poverty

According to the US Census, 18% of Americans are below the poverty line. If I randomly sample $n = 100$ people from the United States, what is the probability that more than 20% of them are below the poverty line?

Example #3: Much More Poverty

According to the US Census, 18% of Americans are below the poverty line. If I randomly sample $n = 1000$ people from the United States, what is the probability that more than 20% of them are below the poverty line?

Example #4: Even More Poverty

According to the US Census, 18% of Americans are below the poverty line. If I randomly sample $n = 10000$ people from the United States, what is the probability that more than 20% of them are below the poverty line?